PegShield

github.com/fexx301/pegshield

Parametric USDC depeg protection, settled by Attestcoin.

A user buys fixed-coverage protection on Creditcoin CC3. When the pinned Ethereum USDC/USD aggregator publishes two qualifying below-threshold updates, anyone submits the two Attestcoin proofs in **one atomic CC3 transaction** — and the pool pays the exact locked coverage to the beneficiary stored at purchase.

The adjuster becomes an on-chain fact.

BUIDL CTC 2026 Fall | Attestcoin Protocol | Cross-chain DeFi | Live on testnet

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THE PROBLEM

A payout decided by an operator has only moved the adjuster problem.

Traditional insurance waits on a human adjuster. On-chain versions replace the adjuster with an oracle operator — who can be **wrong, gamed, or down at the exact moment a claim must settle.** Whoever is waiting for the payout only has the operator's word.

Adjuster

Slow, contestable, opaque. The beneficiary holds a promise, not a fact.

Oracle operator

Centralized assertion. The same trust, relocated — with a nicer API.

Bridge-based cover

Locks assets in vaults: the target of billions in hacks.

THE ANSWER

An oracle can assert a depeg. PegShield verifies the receipt.

> *The payout is decided by the Ethereum receipt of the Chainlink transmission itself — inclusion-proven and continuity-anchored by Attestcoin. No settlement operator can assert, alter, or withhold what the reference feed transmitted.*

Kept

- The industry-grade reference feed (pinned USDC/USD aggregator)
- Fixed coverage and locked beneficiary, chosen at purchase

Removed

- Assertion trust: no one tells the contract what the feed said
- Relayer trust: submitter cannot influence a single field of the decision

The CC3 contract re-derives the answer, round, and timestamp from the authenticated log. The relayer submits opaque proof bytes and nothing else.

HOW IT WORKS

One transaction, every check.

Ethereum		Attestcoin		Adapter		PegShieldPool
Pinned USDC/USD aggregator emits AnswerUpdated	→	Proof service anchors the receipt; BlockProver verifies inclusion + continuity	→	EVM-v1 decoder extracts the exact log; code-hash-pinned	→	Checks both proofs atomically, pays the locked beneficiary

Both proofs settle in a single `submitClaim` — a permissionless relayer cannot pin an unusably late first observation, because a failed or maliciously ordered pair leaves the policy untouched.

- ✓ Receipt succeeded on source chain
- ✓ Positive answer, strictly below trigger
- ✓ Confirmation strictly later (round + time)
- ✓ Per-policy replay protection on attested identity
- ✓ Correct chain key + emitter + event topic
- ✓ Both observations inside the coverage window
- ✓ Minimum breach interval met
- ✓ Effects before transfer; exact token-delta accounting

WHY ATTESTCOIN — INTEGRATION DEPTH

Not a single event read.
A funded, atomic, two-proof settlement.

ATTESTCOIN COMPONENT	WHERE	WHAT IT SETTLES
BlockProver precompile `0x…FD2`	Adapter's only window — `verifySourceLog`	Transaction inclusion (Merkle) + chain continuity, per proof
EVM-v1 decoder `0x731c…` (code-hash pinned)	Same adapter	Receipt status, emitter, exact log at a bounds-checked position
ChainInfo precompile `0x…FD3`	Deployment preflight	Dependency set is live and attesting before any broadcast
Official proof service (batch + single)	Worker CLI + web claim service	Fresh continuity anchors; per-tx single-item paths for distant blocks
Discovery lock	Every consumer	Chain IDs, official addresses, code hashes, event topic — pinned, not caller-configurable

LIVE EVIDENCE — SETTLED 2026-09-12

Same-day purchase → payout: 14 minutes.

08:01
Policy purchased (100 tUSD coverage, 2.5 tUSD premium)

1 tx
Atomic claim: both Attestcoin proofs verified and paid in one transaction

100
tUSD paid to the beneficiary locked at purchase — exact to the wei

FACT	VALUE / RECEIPT
Source rounds (Ethereum)	`1187 (0.99986193, 07:47:15 UTC) + 1188 (0.99986417, 08:40:35 UTC) — exactly 3,600 s apart`
Atomic claim + payout	`creditcoin-testnet.blockscout.com/tx/0x575030ff…c2120c8` · 616,966 gas · 4 events
Accounting	`reserved 1,000,000,000 → 0 · accounted 1,002,500,000 → 902,500,000`
First end-to-end settlement (v3, 2026-09-04)	`Rounds 1171/1172 · 835,156 gas · same exact-payout shape`

ENGINEERING RIGOR

Evidence-first, all the way down.

58 Foundry tests

- 256-run invariant fuzzing (reserved ≤ accounted ≤ balance)
- ERC-20 callback/reentrancy regression
- Named griefing-vector regression: late-first-observation pinning

29 worker + 28 web tests

- Artifact integrity, strict-below trigger boundary, rate limits, 404 mapping
- Playwright e2e: fixture walkthrough, prepare-failure recovery

Gates that bite

- CI green (6 consecutive): contracts, typescript, schemas, e2e
- ABI drift checks parse real TS exports, compare full signatures
- Schema-validated manifests; pinned dependency code hashes

A 28-document evidence chain (P01–P29) records every discovery, deployment, incident, and settlement with receipts — including the failures.

PROTOCOL CONTRIBUTIONS

What we learned about the Attestcoin Protocol.

Proof lifetime

A committed proof survived BlockProver verification **61 hours** after generation. A single-proof continuity path later failed as the attestation window advanced; a fresh batch request restored the same round.

Shared-anchor limits

A two-transaction proof batch shares one continuity anchor — valid for nearby headers only. For distant blocks, the later proof fails with **Merkle root mismatch**; per-transaction single-item batches verify. Confirmed at the BlockProver level.

CC3 estimateGas

`eth_estimateGas` is intermittently unstable (observed ~50% of actual need — a broadcast ran out of gas at 110,352 vs a fresh estimate of 226,245). Broadcast with explicit gas limits.

Operational rules any Attestcoin builder needs — documented with receipts, published back to the ecosystem.

HONESTY AS A FEATURE

The boundary, stated precisely.

Removed

- Settlement-layer operators: no one between the Ethereum receipt and the CC3 payout can assert, alter, or withhold
- Relayer, browser, and server trust
- Caller-supplied price, round, timestamp, beneficiary, feed, or hash

Retained — by design

- Reference-feed quality: a wrong aggregator answer still pays, exactly as real parametric products insure the index, not ground truth
- Aggregation between the two observations is not proven uninterrupted

> *Feed choice and window are product parameters pinned per policy at purchase — documented, not hidden. That is the difference between a trust boundary and a trust claim.*

CROSS-CHAIN DEFI

Cover the collateral, unlock the credit.

A CC3 lending market names itself as the policy beneficiary at purchase. Depeg cover on the USDC collateral is what lets undercollateralized lending against it survive the tail event — protection settles natively where the credit application needs liquidity.

Product

- Fixed coverage, premium in bps (ceil-rounded on-chain)
- Underwriter-funded pool, exact reserve accounting
- Immutable per-policy terms; expiry; one-way product disable

Stack

- Contracts: pool + adapter (Foundry, 0.8.28)
- Worker: discovery, proof build, claim calldata CLI
- Web: wallet-gated dashboard + public discovery API

Verify everything

- github.com/fexx301/pegshield
- README → "Why Attestcoin" + verify-everything table
- docs/evidence/P29 — same-day settlement with receipts

Buy fixed coverage. Prove the depeg twice. Let Creditcoin pay.